Prathamesh Patil

Software Engineer · Full Stack & AI

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Summary

B.E. in AI & ML (Oct 2026) with 11 months of internship experience across backend services and LLM applications. I build products end to end, from database design and APIs to the interface, and from data and model training to evaluation, and I measure what I build before deciding what to ship.

Experience

AI & Cloud Computing Intern, SuprMentr Technology Pvt Ltd — Nov 2025 – Mar 2026

- Worked on a multi-agent RAG platform built with LangGraph, including a self-evaluation step that tracks hallucination rate and answer accuracy.
- Wrote SQL (window functions, CTEs, aggregations) over PostgreSQL to produce weekly usage analytics shown in React dashboards.

Software Engineer Intern, AI Division, Jmedia Corp — Jan 2025 – Jun 2025

- Built REST APIs with FastAPI that serve LLM-backed features, with JWT authentication and rate limiting.
- Containerized services with Docker, automated build and deployment with GitHub Actions, and monitored latency and error rates on Grafana dashboards.

Projects

Slotwise: Event Booking Platform — github.com/Prathameshh-Patil/slotwise

FastAPI, PostgreSQL, Celery, Redis, Next.js, Docker

- Built a full-stack booking product with a Next.js frontend, a FastAPI backend, and background jobs for confirmations.
- Prevented double booking with a partial unique index and all-or-nothing transactions in PostgreSQL, verified by a test that fires simultaneous requests at one seat and asserts exactly one succeeds.

IntentBench: Three Ways to Classify Text, Measured — github.com/Prathameshh-Patil/intentbench

Python, pandas, scikit-learn, PyTorch, Hugging Face, FastAPI

- Trained and compared TF-IDF with logistic regression, a fine-tuned DistilBERT, and zero-/few-shot LLM classification on 150 intents (CLINC150), tuning only on validation and touching the test set once.
- Reached **91.8% accuracy and 0.939 macro-F1** with DistilBERT plus a validation-chosen confidence threshold, which raised out-of-scope recall from 0.59 to 0.78.
- Served the model behind FastAPI at **12 ms median latency** vs 1.6 s for the LLM, with no per-request cost.

ReplyDesk: AI Support Inbox — github.com/Prathameshh-Patil/replydesk

Python, FastAPI, PostgreSQL, LLM API, Next.js

- Built four prompt-driven agents that classify a customer message, extract entities, draft a reply, and check it against guidelines; a human approves every reply before it is sent.
- Wrote an evaluation script over 60 labelled messages and used it to revise prompts: category accuracy rose **90% to 97%**, entity extraction held at 98%, and no fields were invented.

Loan Document Underwriting Agent

Python, Lyzr Agent Studio, Tesseract OCR

- Built a 12-agent pipeline that extracts and cross-checks mixed loan documents into one summary with confidence scores and risk flags, with deterministic validators for PAN, Aadhaar, GSTIN, and IFSC.

Skills

Languages: Python, SQL, JavaScript, TypeScript
Backend: FastAPI, Node.js, REST APIs, JWT authentication, PostgreSQL, SQLAlchemy, database transactions
Frontend: React, Next.js
AI / ML: PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy, model evaluation, LLM APIs (Gemini, Groq), RAG, LangGraph, multi-agent pipelines
Tools: Docker, GitHub Actions CI/CD, Git, Grafana

Education

B.E., Artificial Intelligence & Machine Learning — 2022 – 2026
Dayananda Sagar Academy of Technology and Management, Bengaluru — CGPA 7.6